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Statistical Analysis of Type 2 Diabetes

Exploratory, Inferential and Predictive Modelling Using the Pima Indians Dataset

MSc in Artificial Intelligence
Numerical Analysis
January 2026

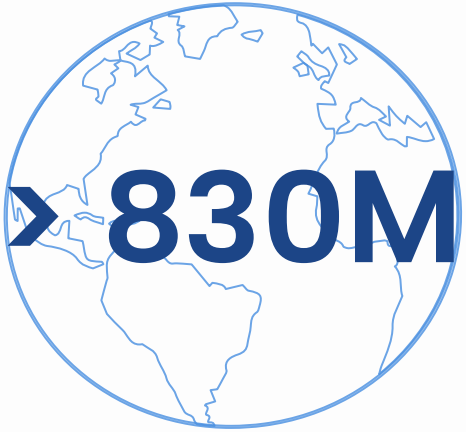
The Global Diabetes Challenge

Chronic metabolic disease → persistent hyperglycaemia

Major cause of multi-organ damage:

- Cardiovascular System
- Kidneys
- Eyes
- Nerves

Many complications are preventable → need for early detection & management



(World Health Organization, 2024)

Focus on Type 2 Diabetes & Risk Factors

> 90% → Type 2 Diabetes → insulin resistance / inadequate insulin utilisation

(International Diabetes Federation, 2024)

Risk Factors

- Overweight/Obesity
- Physical Inactivity
- Unhealthy diet
- Increasing age
- Metabolic factors

(International Diabetes Federation, 2024)

Clinical predictors

- Blood Glucose Concentration
- Body Mass Index (BMI)
- Blood Pressure
- Age

(American Diabetes Associatio 026)

Research Aims & Dataset

Dataset: Pima Indians Diabetes (Kahn, n.d.)

Sample	768 adult women
Clinical Predictors	○ Glucose (Continuous) ○ BMI (Continuous) ○ Blood Pressure (Continuous)
Demographic	○ Age (Continuous) ○ Number of pregnancies (Integer)
Outcome	○ Binary Diabetes Status Positive / Negative

Aims

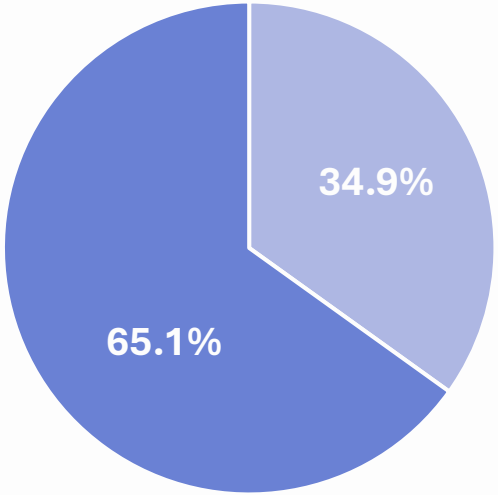
1. **EXPLORE**
2. **TEST**
3. **DEVELOP**

01

Data exploration & Descriptive Statistics

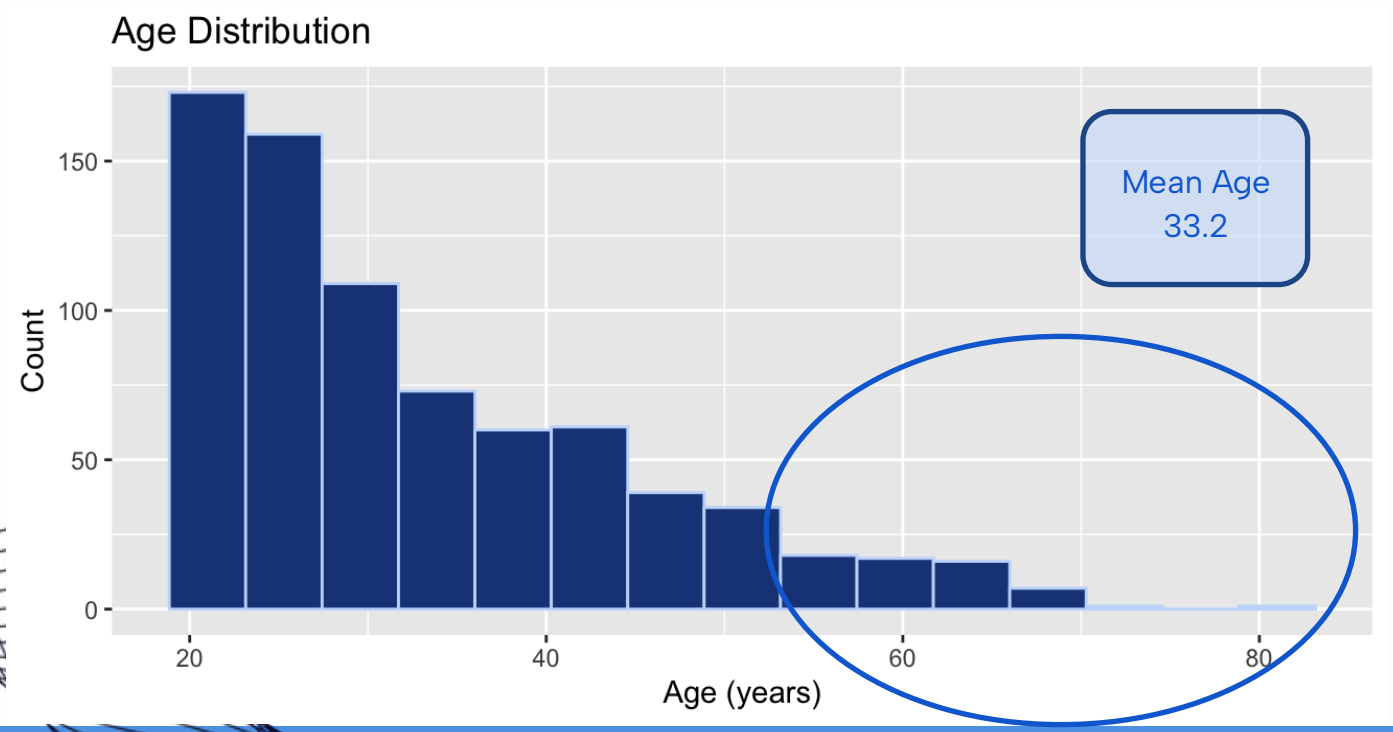
Cohort Description, Data Assessment, and Risk Factor Distributions

Exploratory Data Analysis – Cohort Overview



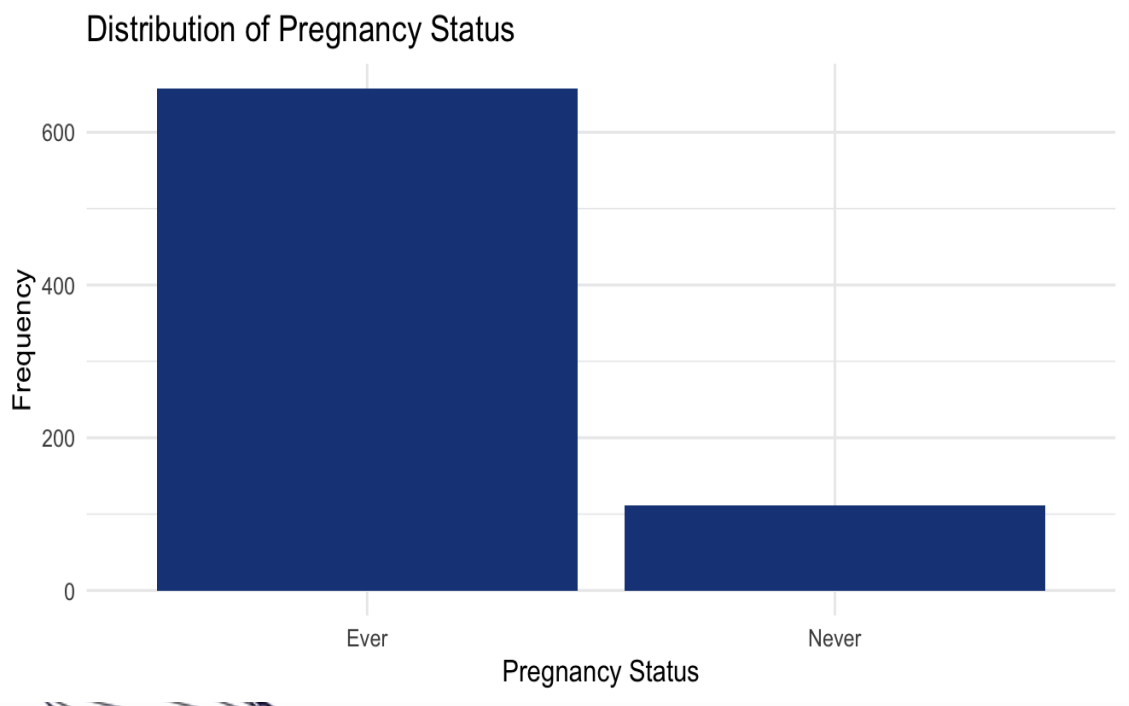
2:1 class imbalance

Participant Characteristics: Age



Distribution is skewed to the right

Participant Characteristics: Pregnancy History



Pregnancy categories

Group	Number	Percentage
0	111	14.45%
1 - 2	238	30.99%
3 - 4	143	18.62%
5+	276	35.94%

Clinical Profile & Data Quality Note

Variable	Mean	SD	Median	Min	Max	Skew
Age	33.24		29	21	81	1.13
BMI	**31.99**	7.88	32	0*	67.1	-0.43
Glucose	120.89	31.97	117	0*	199.0	0.17
Blood Pressure	69.11	19.36	72	0*	122.0	-1.84
Pregnancies	3.85	3.37	3	0	17.0	0.90

Obesity:
BMI ≥ 30

(World Health Organization, 2025)

Clinical Profile & Data Quality Note

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Normal range

(John William McEvoy et al., 2024)

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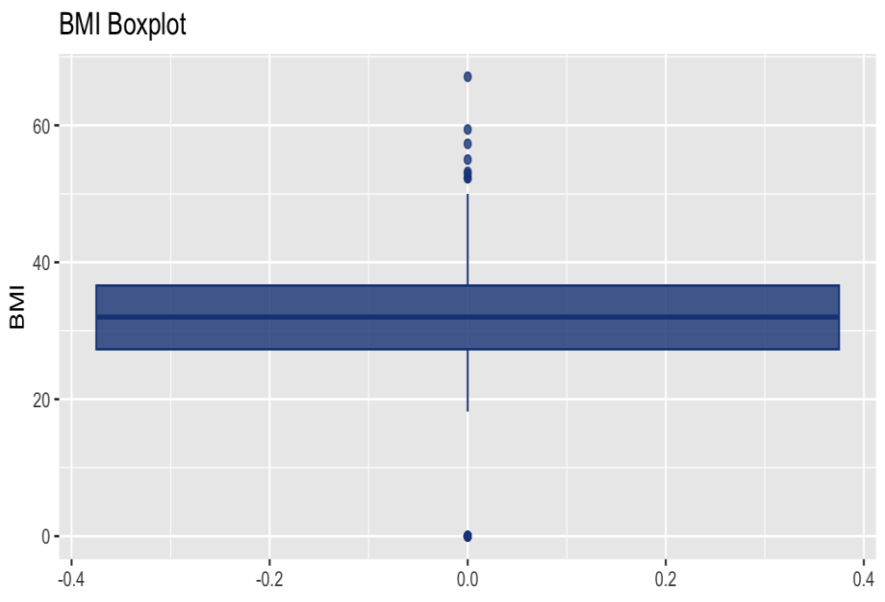
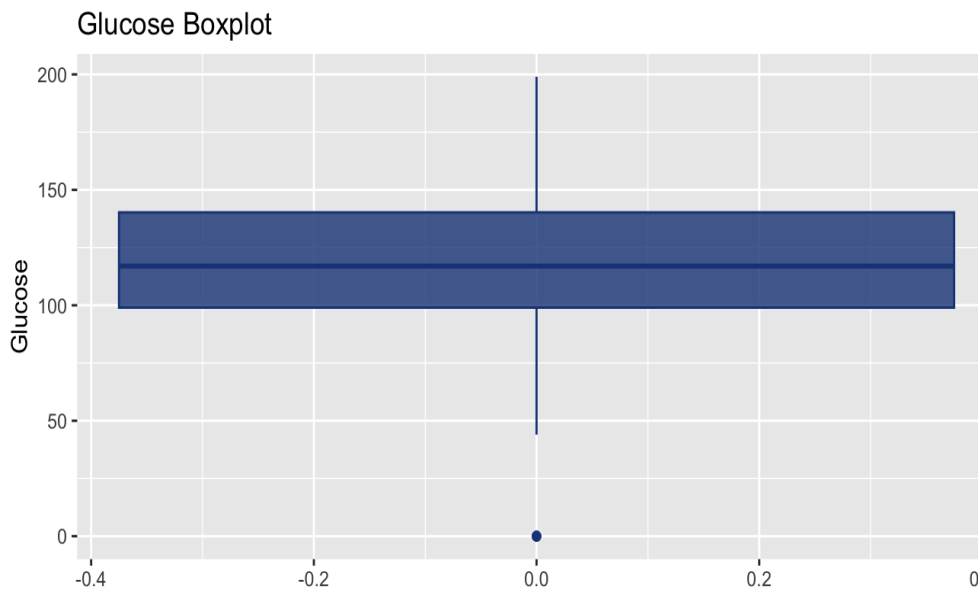
Note: *Zeros likely represent missing data rather than true physiological zeros.

Visualising Variation & Outliers





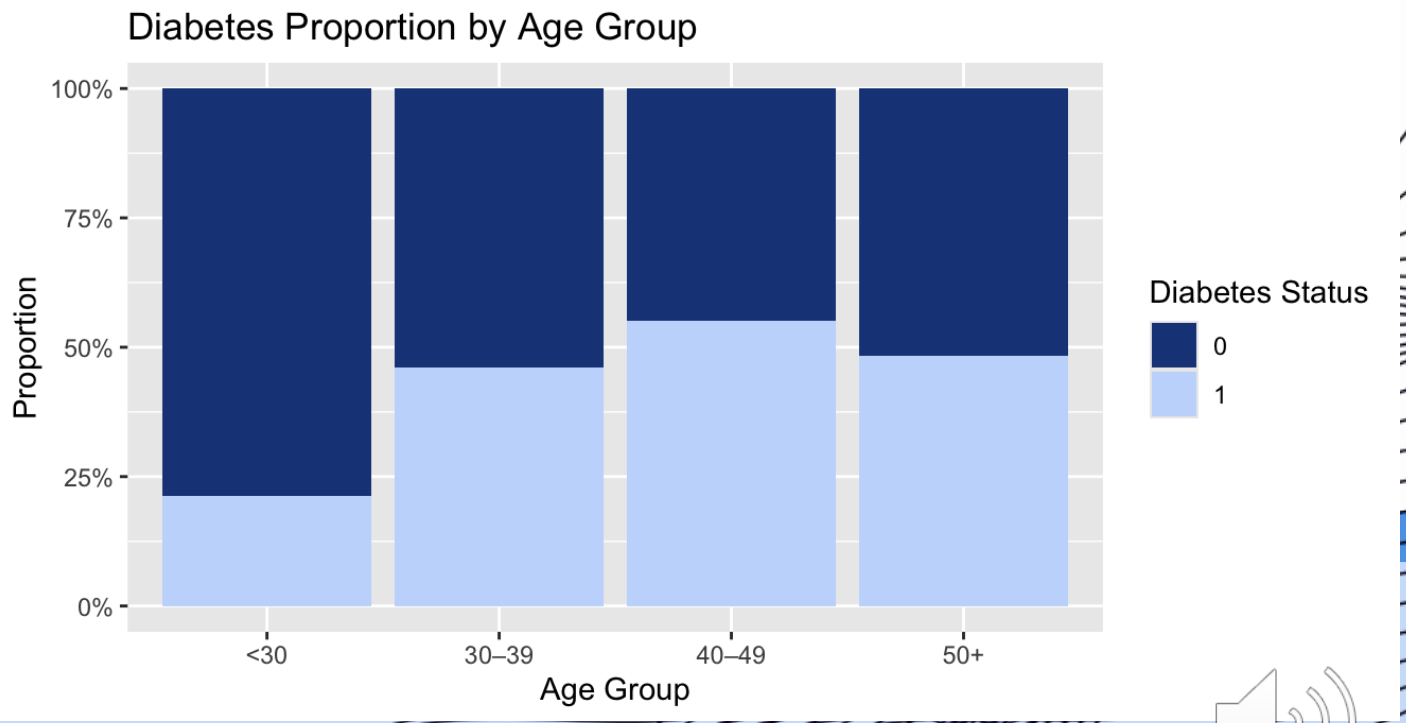
- Outliers > 40 → Obesity cases
- Zeros at 0 → Missing data

Diabetes Prevalence Across Key Categories: Age Group

KEY INSIGHTS

- 20s → ~ 21%
- 30 - 39 → ~ 46%
- 40 - 49 → ~ 55%
- 50+→ ~ 48%

Peak prevalence

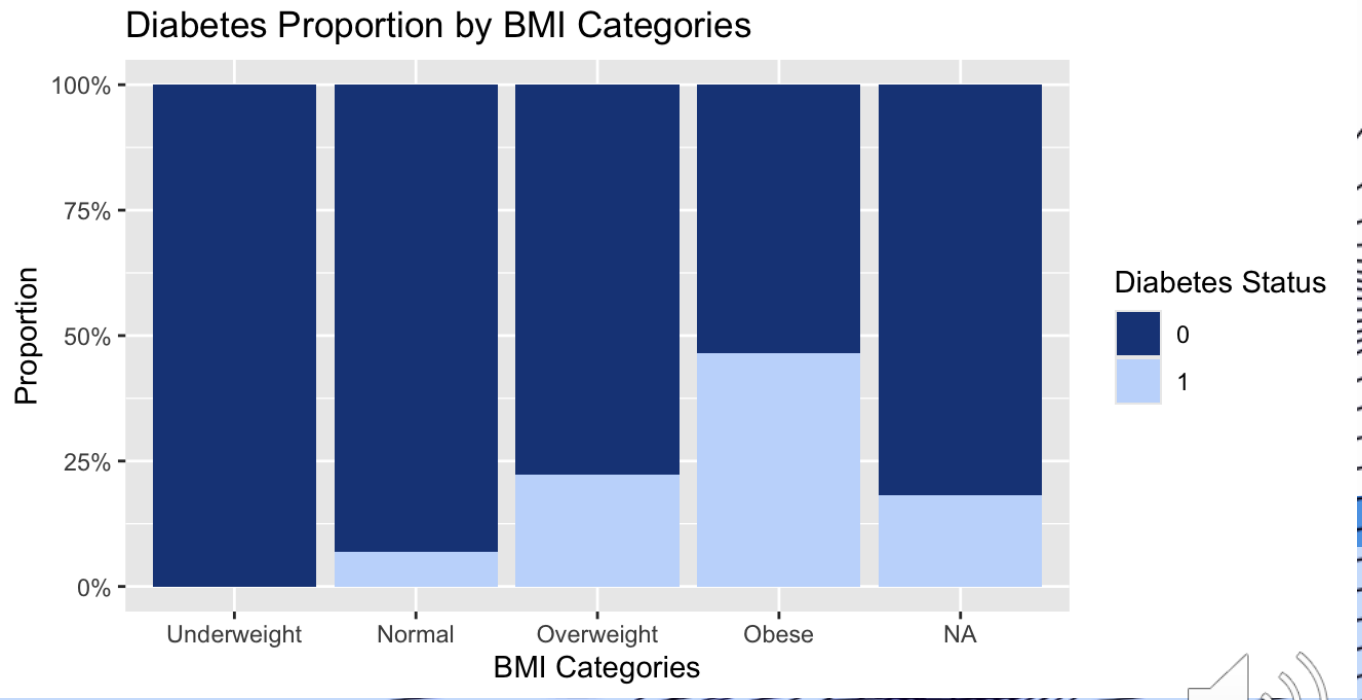


Diabetes Prevalence Across Key Categories: BMI Categories

KEY INSIGHTS

- Underweight → 0%
- Normal → ~ 7%
- Overweight → ~ 22%
- Obese → ~ 46%
- NA → ~ 18%

Peak prevalence

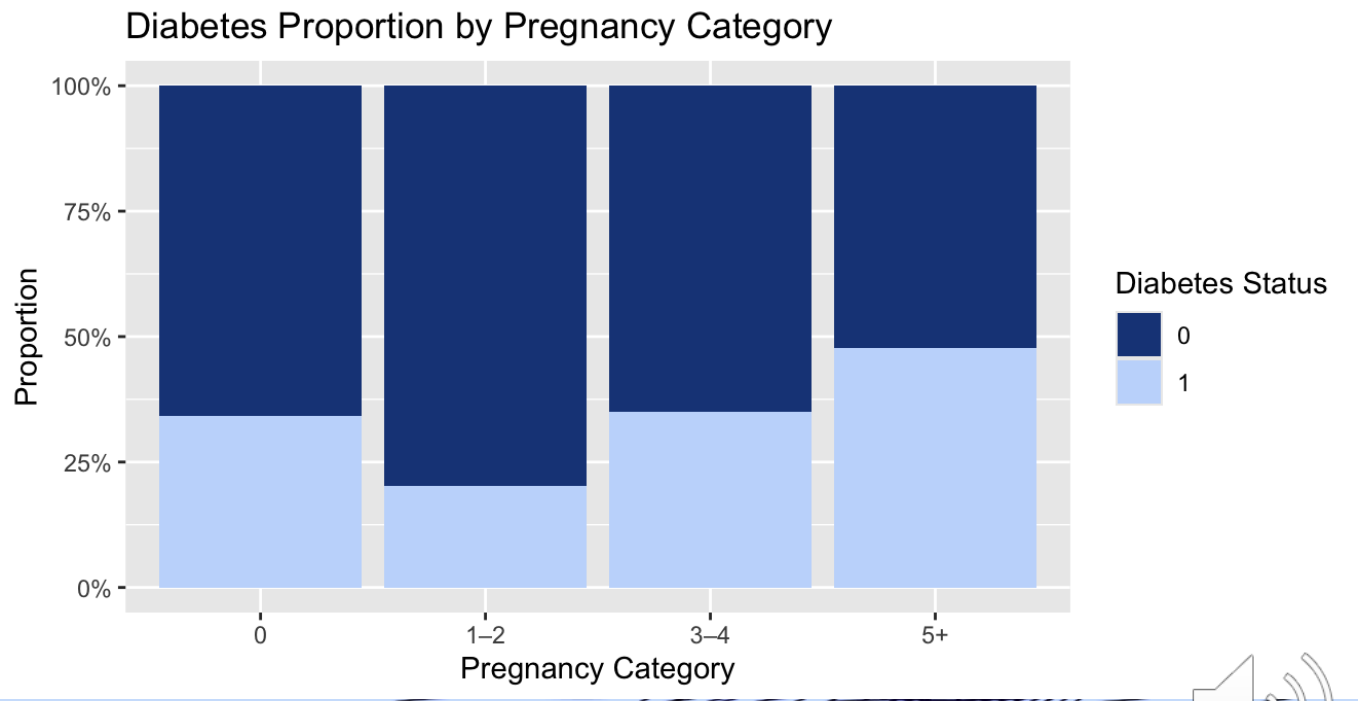


Diabetes Prevalence Across Key Categories: Pregnancy Categories

KEY INSIGHTS

- 0 → ~ 34%
- 1 – 2 → ~ 20%
- 3 - 4 → ~ 35%
- 5+→ ~ 48%

Peak prevalence

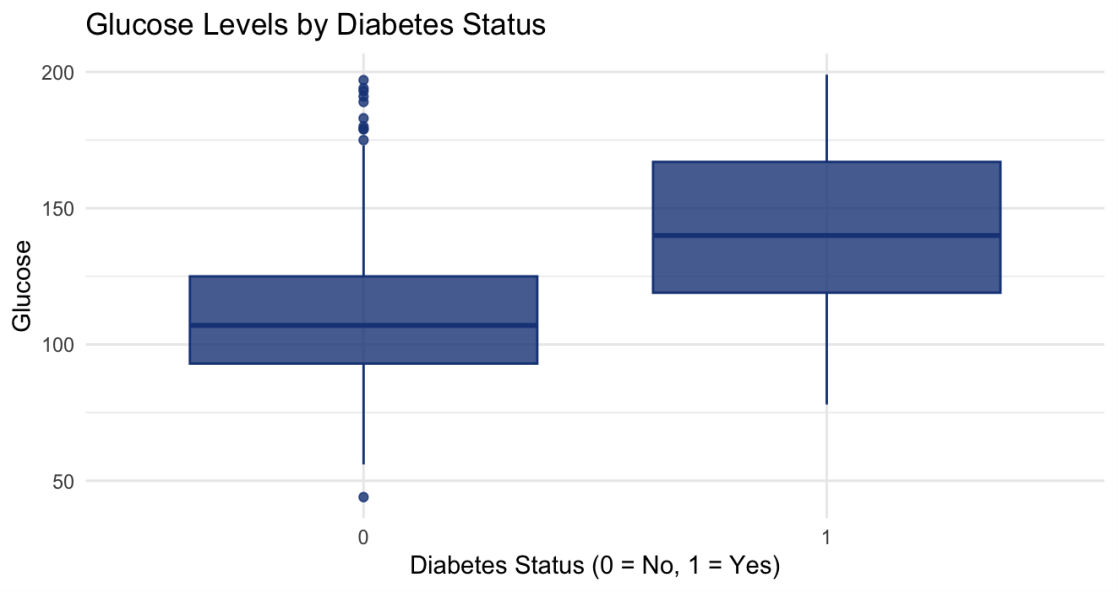


02

Inferential Statistics

Hypothesis Testing for Group Differences and Risk Factor Associations

Group Differences: Glucose



Shapiro-Wilk test

Result:

Non-diabetic: p-value = $5.447 \times 10^{-9} < 0.001$

Diabetic: p-value = $6.587 \times 10^{-7} < 0.001$

Welch test

Result: $t(461.33) = -13.75$, $p < 2.2 \times 10^{-16}$

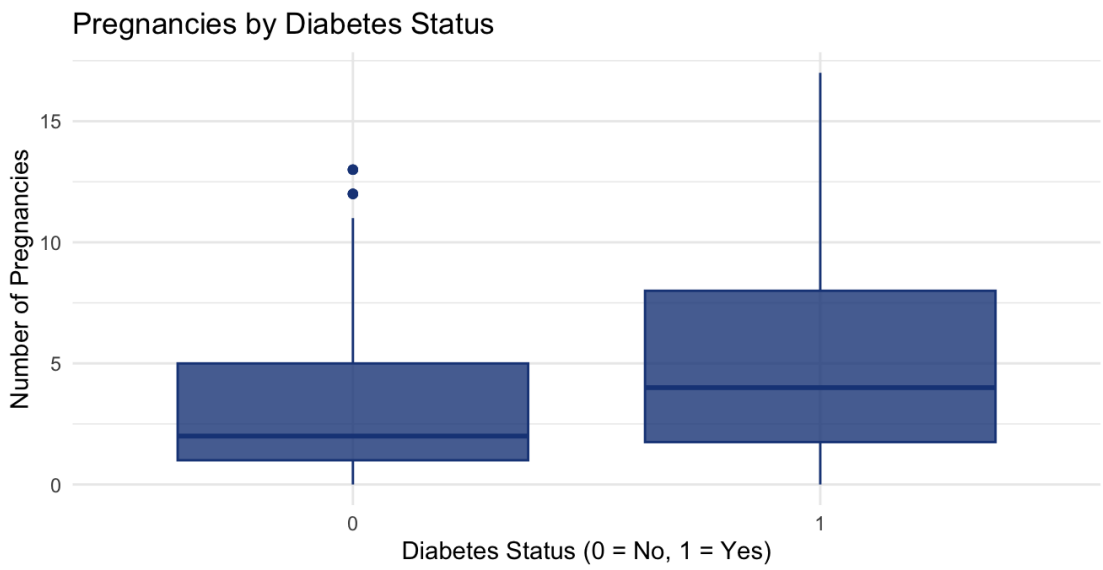
Shapiro-Wilk Test Hypotheses

H_0: Mean glucose is equal in diabetic and non-diabetic groups.

H_1: Mean glucose differs between groups.

Group	Mean (95% CI)
Non -diabetic	109.98
Diabetic	141.26

Group Differences: Pregnancies



Shapiro-Wilk test

Result:

Non-diabetic: p-value < 2.2×10^{-16}

Diabetic: p-value < 1.157×10^{-8}

Wilcoxon test

Result: W = 50.985, p-value = 3.745×10^{-8}

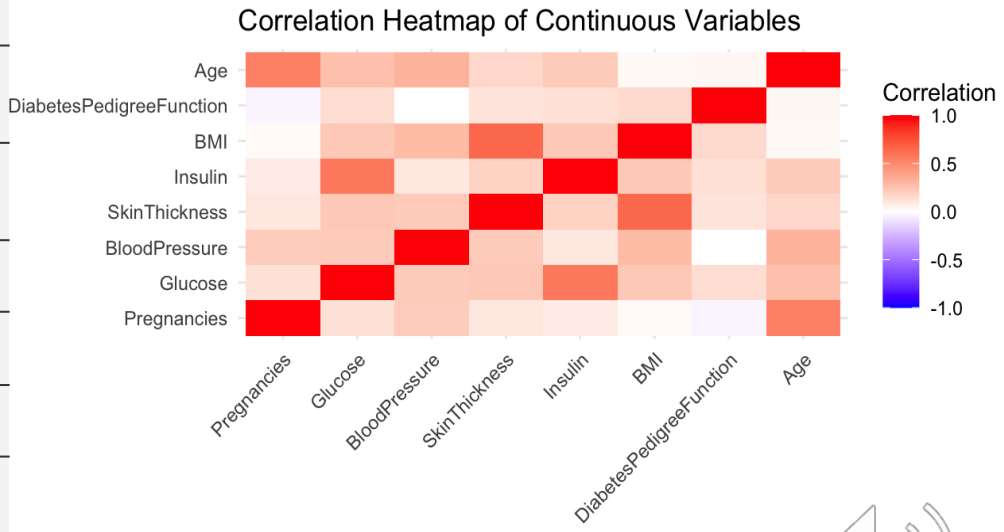
Shapiro-Wilk Test Hypotheses

H_0: Pregnancy distributions are the same in diabetic and non-diabetic groups.
H_1: Pregnancy distributions differ between diabetic and non-diabetic groups.

Correlation Matrix

	Preg	Gluc	BP	Skin	Ins	BMI	DPF	Age
Pregnancies	1.0	0.13	0.21	0.10	0.08	0.02	−0.03	**0.54**
Glucose	0.13	1.0	0.22	0.23	**0.58**	0.23	0.14	0.27
Blood Pressure	0.21	0.22	1.0	0.23	0.10	0.29	0.00	**0.33**
Skin Thickness	0.10	0.23	0.23	1.0	0.18	**0.65**	0.12	0.17
Insulin	0.08	**0.58**	0.10	0.18	1.0	0.23	0.13	0.22
BMI	0.02	0.23	0.29	**0.65**	0.23	1.0	0.16	0.03
DPF	−0.03	0.14	0.00	0.12	0.13	0.16	1.0	0.03
Age	**0.54**	0.27	**0.33**	0.17	0.22	0.03	0.03	1.0

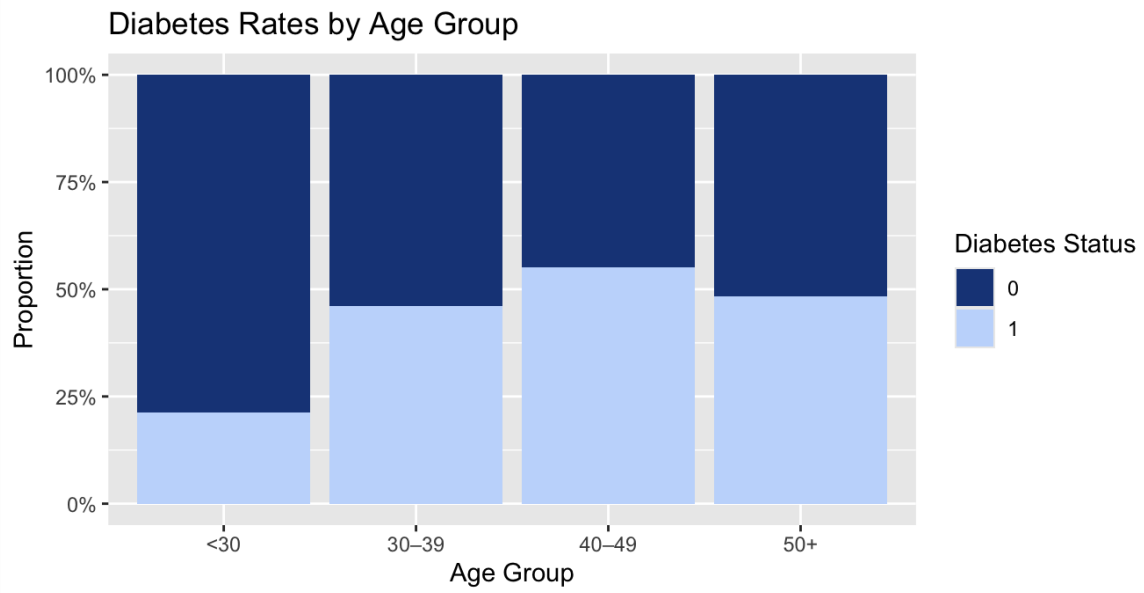
Potential multicollinearity



Categorical Associations: Age Group x Diabetes

Chi-square test

Result: $\chi^2(3) = 69.92$, $p = 4.45 \times 10^{-15}$

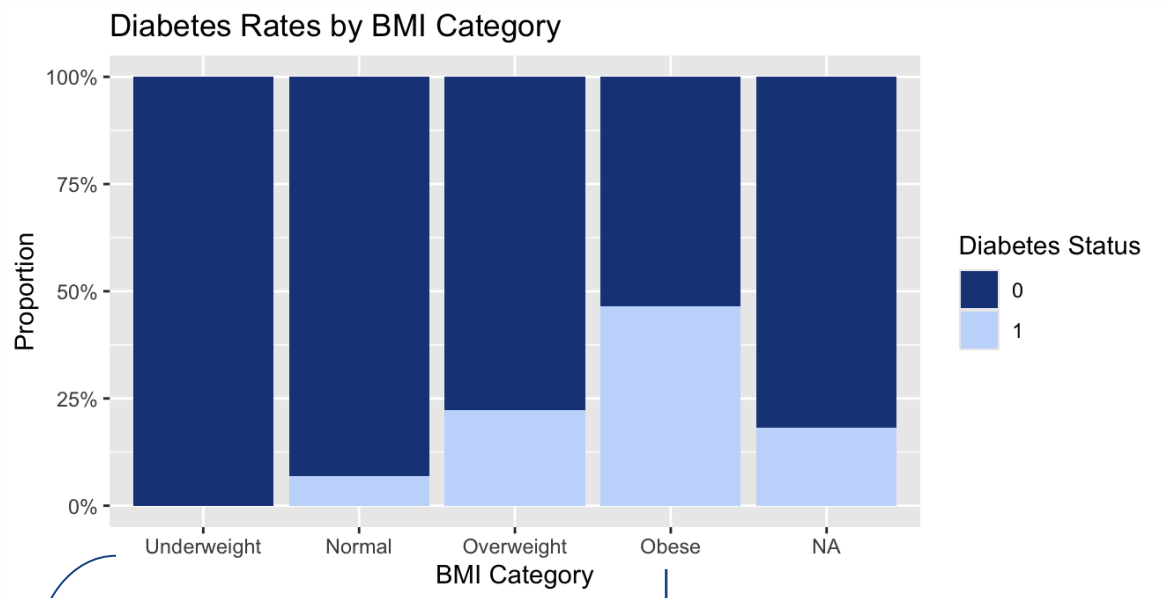


Proportion of diabetes increases with age

Categorical Associations: BMI Category x Diabetes

Fisher's Exact test

Result: $p < 2.2 \times 10^{-16}$



no values

most common

Mean Glucose Across Age Groups

One-Way ANOVA

Result: $F(3, 759) = 20.44$, $p = 9.57 \times 10^{-13}$

Age Group	Sample Size (n)	Mean Glucose	vs. < 30	vs. 30 - 39	vs. 40 - 49
< 30 years	268	105.0	-	Higher	Higher
30 - 39 years	176	116.5	Higher	-	Same
40 - 49 years	140	116.1	Higher	Same	-
50+ years	180	129.9	Highest (+24.9)	Higher	Higher

Significance Key:

- Higher: Significantly higher glucose ($p < 0.01$)
- Highest: Largest difference (~25 units higher than <30)
- Same: No significant difference ($p > 0.05$)

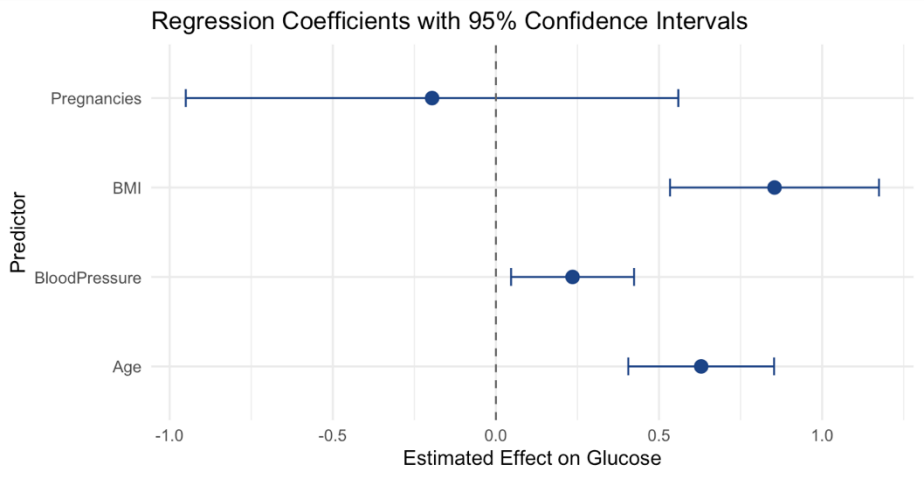
03

Regression Analysis

Linear and Logistic Regression for Risk Prediction and Model Assessment

Linear Regression: Model specifications & Results

Model fit: $F(4, 719) = 25.54$, $p < 0.001$; Adjusted $R^2 = 0.119$



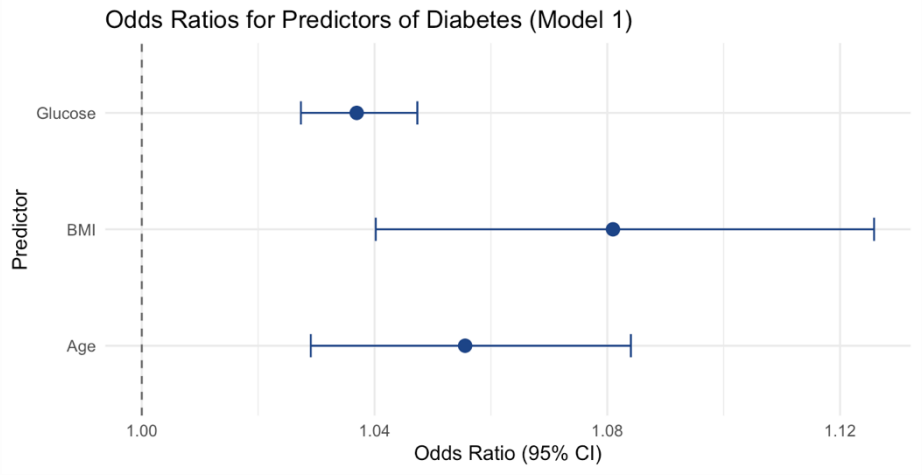
Predictor	Estimate	95% CI	p-value
Age	0.63	[0.41, 0.85]	< 0.001
BMI	0.85	[0.53, 1.17]	< 0.001
Blood Pressure	0.23	[0.05, 0.42]	0.015
Pregnancies	-0.20	[-0.95, 0.56]	0.611

Logistic Regression Models: Model 1

Model Overview

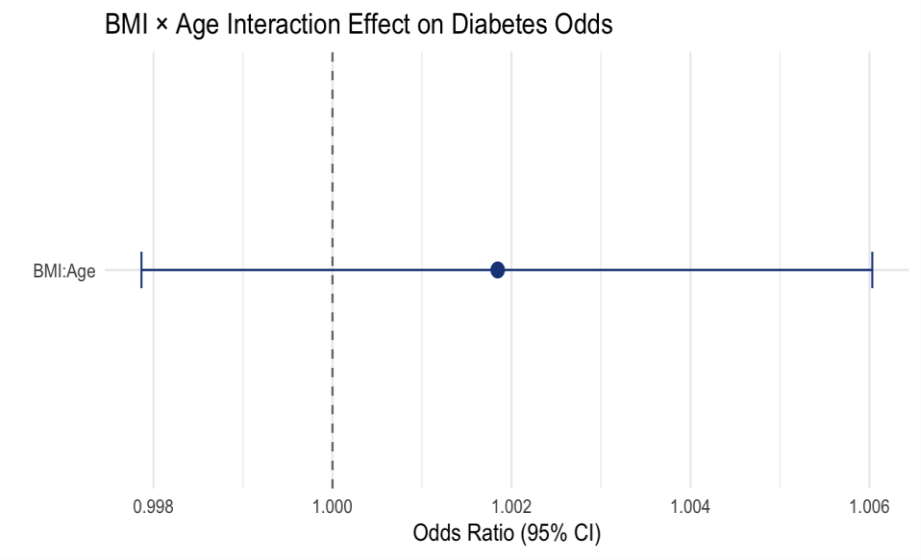
Predictors: BMI + Age + Glucose

Binary Outcome: Diabetes status



Predictor	OR	95% CI	%Change in Odds
BMI	1.08	[1.04, 1.13]	+8.1%
Age	1.06	[1.03, 1.08]	+5.6%
Glucose	1.04	[1.03, 1.05]	+3.7%

Logistic Regression Models: Model 2



Model Overview

Predictors: BMI, Age, Glucose, BMIxAge

Binary Outcome: Diabetes status

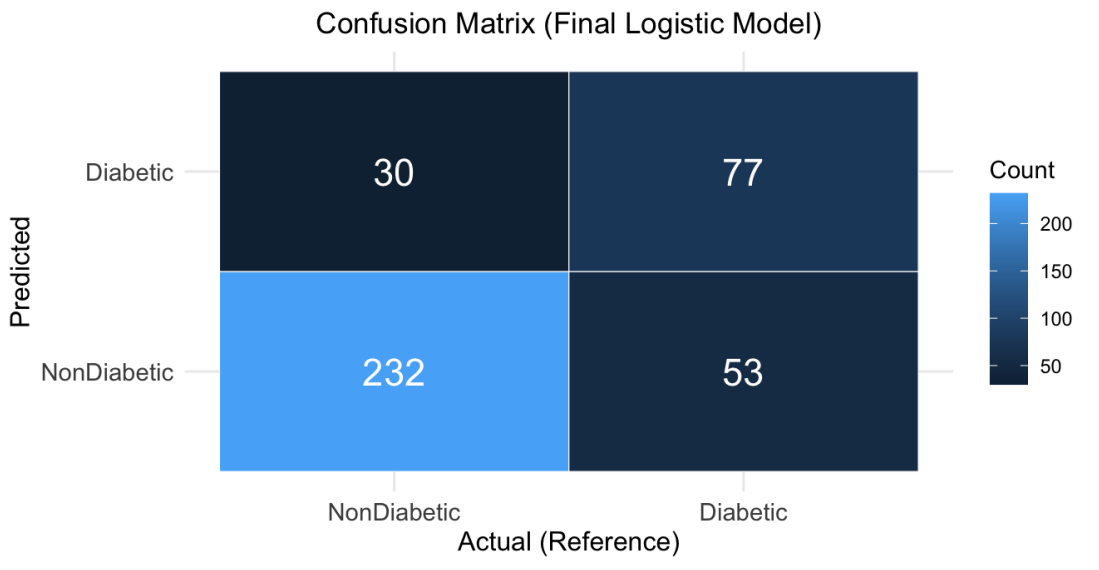
No evidence that BMI's effect differs by age

Model Comparison & Selection

Criterion	Model 1	Model 2
Interaction	No	Not significant
p-value	-	0.37
AIC	362.4	363.6
Multicollinearity	Low	High (VIF > 10) Age and BMIxAge

Preferred model:
Model 1

Predictive Performance of the Final Logistic Regression



- **Accuracy:** 78.8%
- **Specificity:** 88.6%
- **Sensitivity:** 59.2%

Hosmer–Lemeshow test

p = 0.93 → good model calibration

Conclusions

01

Group differences

Glucose and pregnancies differ by diabetes status

02

Associations

Age and BMI strongly linked to diabetes

03

Regression Models

Glucose, BMI and age are independent predictors

04

Model Performance

Good specificity, moderate sensitivity

Implications for prevention

Healthy BMI – Early glucose monitoring – Target older adults

(Galaviz et al., 2019)

Limitations & Recommendations

Data and Design Considerations

- Cross-sectional data
- No lifestyle variables
- No causal inference

Implications and Next Steps

- Earlier screening
- Targeted prevention
- Adopt prospective studies with follow-up over time
- Include lifestyle & genetics

References

All analyses and visualisations were produced using R statistical software (R Core Team, 2025).

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